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# Barriers to equitable disaster recovery: A scoping literature review

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ABSTRACT

Historically marginalized communities continue to face several barriers when trying to access disaster finances and resources. This scoping review aims to synthesize the existing literature in order to clarify the barriers populations, specifically Black, Indigenous, People of Color (BIPOC), Latine, limited English proficiency (LEP), differently abled, and undocumented populations, face during the disaster recovery process in the United States. Members of the research team independently reviewed the literature (n = 108 articles) and used thematic analysis to find, examine, interpret, and report repeated patterns. Five main interconnected themes emerged with high inter-rater reliability throughout the analysis, which were: 1) the process is confusing and difficult to navigate; 2) exclusion from or lack of access to political power and recovery decision-making processes; 3) exclusion due to discrimination in its many forms; 4) issues in the funding aid ecosystem; and 5) politicians and media having an inordinate amount of control over the flow of resources and recovery processes. Findings point to a need for greater inclusion of historically marginalized populations throughout the pre-disaster planning and recovery process. These findings provide the basis for actionable recommendations for governments, relief organizations, and other stakeholders on how to address recovery barriers and meet the needs of diverse populations.

## 1. Introduction

Disasters are increasing in frequency and intensity across the globe due to anthropogenic climate change [1]. These disasters are disproportionately impacting marginalized communities due to generations of systemic racism and oppression, which have forced many of these communities into hazard-prone areas, further exacerbating the precarity of their lives [2,3]. Despite the ever-increasing

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number and size of disasters and their disproportionate impact on communities with many pre-existing vulnerabilities, it is often these same communities that face the most barriers to recovery.

The increase in climate-related hazards and the continued marginalization of Black, Indigenous, People of Color (BIPOC), Latine, differently-abled, and those with limited English proficiency creates greater vulnerability in the face of said hazards [4–6]. Despite prolonged legal pushback and community activism efforts to address issues around equity and efficiency, the outcomes of emergency management programs continue to differ from those laid out by the equity requirements in the Stafford Act [7–9]. In actuality, disasters and the ensuing responses to them frequently serve to further entrench discriminatory social mechanisms rather than undo them. Disaster recovery programs that are meant to support communities at times of crisis and instability play a major role in exacerbating pre-existing social inequalities through systemic biases [10–12].

Systemic biases are consistently cited in the literature, with these being biases or inequities in outcomes that occur repeatedly or consistently, which may or may not have a clearly identifiable cause, but which are likely due to conscious or unconscious biases, including prejudices and forms of racism [13,14]. Domingue and Emrich [15] suggest that funding distribution is affected by social vulnerability indicators that do not always align with equitable practices. Muñoz and Tate [16] highlight the procedural inequities surrounding the distribution of federal funding that give preference to those with privileged conditions (i.e., those with resources to fund their own recovery) over those in vulnerable conditions, namely elderly, Black, Hispanic, and Indigenous populations, those in nursing homes, and those experiencing unemployment [15]. This results in socially vulnerable populations receiving rejected applications at disproportionate rates, ultimately slowing, or permanently impeding their recovery [8,17].

These systemic biases appear to be a byproduct of an unexamined self and systems, including conscious or unconscious biases at individual levels that become institutionalized through various channels. A few of these channels by which systemic biases are institutionalized include program/procedural design [14], lack of diversity in emergency management staff [5,18,19], and failure to consciously direct resource flows where they are most needed [9,16,17,20,21]. These biases, despite consistent, clearly demonstrable evidence that they exist, appear to frequently go undetected or uncorrected due to the challenges of demonstrating discrimination based on a protected class and the legal system's limited ability to require corrective or punitive measures.

While critiques of programs and outcomes are plentiful, acknowledging when programs have failed is only the first step in rectifying this persistent problem. Understanding where barriers exist, who they are impacting, and what is maintaining them is crucial for dismantling recovery obstacles. Generally, the literature largely discusses and explores barriers in silos by looking at specific programs, disaster types, or specific events, which for depth of analysis is crucial, but it is equally important to step back and take a look at the bigger picture. By broadening the lens, overarching themes and common denominators across programs and responses can be identified, providing valuable insights for more substantive and comprehensive interventions. Some literature has begun to explore barriers to recovery in this way, including Rouhanizadeh et al. [22], which explored barriers to recovery using a global lens and identified 63 barriers, which they summarize in five different categories. More closely related to our review is Wilson et al. [23], which explored flood recovery outcomes and barriers to disaster assistance for vulnerable populations through an equity lens.

Given the frequency of changes in U.S. administrations and the subsequent initiatives, most recently, ones such as the Justice40 Initiative and the launch of the Building Resilience Infrastructure and Communities (BRIC) program, with its goal to prioritize the most vulnerable communities, further analysis of past responses, outcomes, and programs is warranted. There is an opportunity to capitalize on the current momentum and further elucidate historic issues and how they could be addressed. While there are several existing reviews of disaster recovery outcomes, this review focuses on a broader temporal range of literature and focuses on historically marginalized groups in the United States and the barriers they face to accessing disaster recovery resources. This literature review strives to further examine the root causes of these recovery barriers for communities in the United States and its territories. Furthering our understanding of disparities in recovery outcomes can then inform initiatives to help communities avoid the downward spirals of accelerated resource losses and further marginalization often associated with disasters [24,25].

Given the increasing number of disaster events, their impact, and the continued call for equitable outcomes in disaster recovery, exploration of challenges faced by communities and support programs is needed. The purpose of this scoping review is to identify and present themes from the past 25 years of research and practice on this subject. Specifically, the aim of this scoping review is to explore, through an equity lens, the barriers communities (particularly historically marginalized communities) face when navigating access to disaster financing and resources, with particular focus on those barriers that are not unique to any specific program or response.

## 2. Methodology

A scoping literature review of the last 25 years was carried out from June 2022 to March 2023, predominantly using EBSCOHost and Google Scholar search engines (see Fig. 1 for an overview). While there are many types of reviews [26,27], a scoping review was selected because this method aligns well with our aim for this study as scoping reviews are used to "identify knowledge gaps, set research agendas, and identify implications for decision-making" [28, p. 1]. The basic steps laid out in the literature for conducting rigorous scoping reviews were followed [29,30]. The research team started the search with a boolean search including the key terms "barrier", "disaster", and "recovery" to identify a broad range of relevant articles from various contexts. The research question that

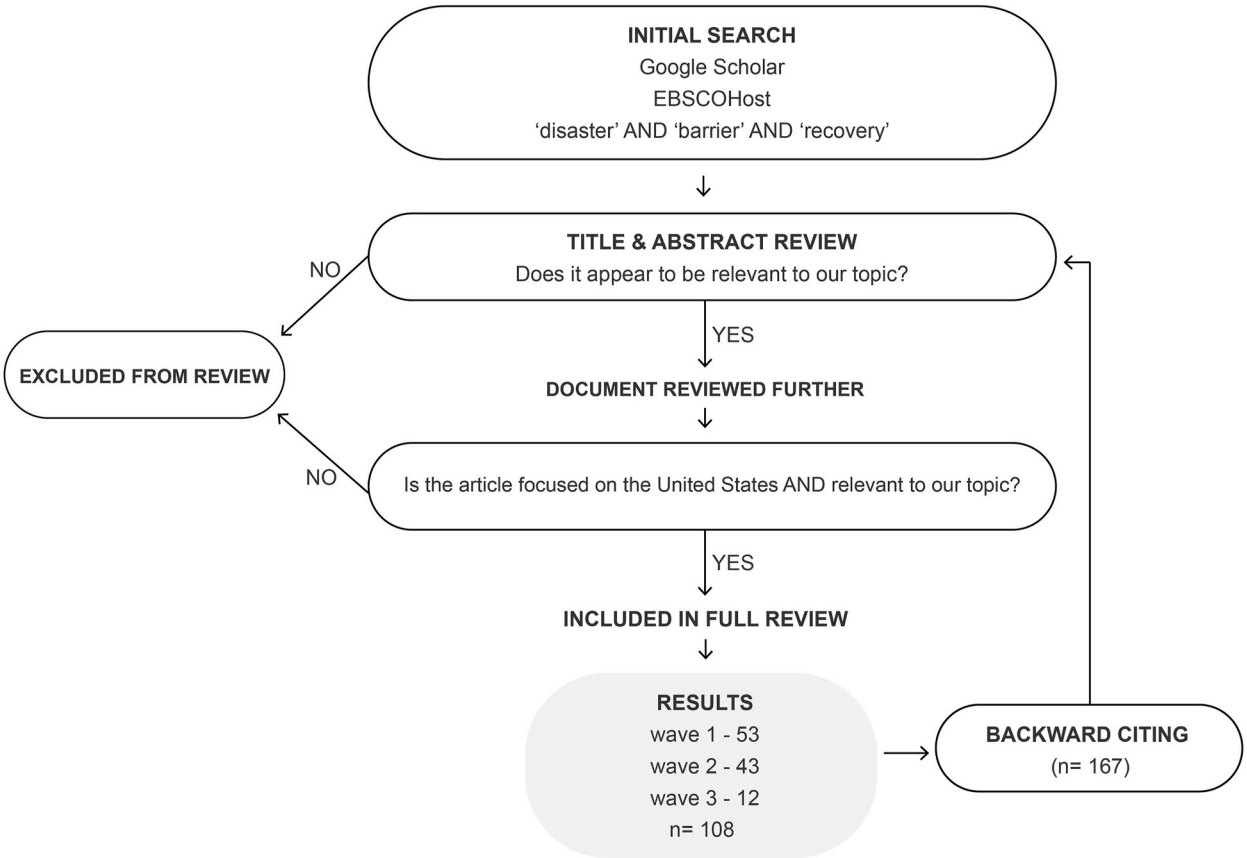


**Fig. 1.** Literature review process flow chart.

guided this review was "*what barriers to disaster recovery do historically marginalized populations face when trying to access disaster finances and resources in the United States*".

### 2.1. Article screening and selection

A group of researchers reviewed titles and abstracts for relevance and to ensure they met inclusion criteria, which were: (a) the article was relevant to the research question; (b) it was focused on the United States; and (c) the article was available in English. If the articles met the inclusion criteria, they were included for further review. Articles meeting inclusion criteria were imported into a shared group Zotero library, where they were carefully screened and annotated using a shared color-coding system. As the researchers annotated the articles from the first wave of searches (n = 53), they simultaneously noted other relevant citations through backward citation tracking, more commonly known as snowball sampling of references, which has been demonstrated to be an effective tool for identifying relevant articles for inclusion [31,32]. This snowball sampling method yielded an additional 167 potentially relevant citations, which were further reviewed in a second wave of analysis. Only 43 of these met the inclusion criteria and were included in the study. Finally, an additional wave of searches was conducted to ensure the most recent literature had been captured, which yielded an additional 12 articles that met inclusion criteria for a total of 108 articles that were analyzed for the purpose of this study.

### 2.2. Data analysis

Each article that met inclusion criteria was then analyzed for relevant content. The model of thematic analysis suggested by Miles and Huberman [33] guided our methodological approach with the basic components of data reduction, data display, and drawing conclusions. The same methods were used for each of the three waves, and therefore are only outlined once here. Data reduction was completed, as Miles and Huberman [33] suggest, through sorting, discarding, and organizing carefully selected snippets of text as well as the summarization of key passages. This process of data reduction was completed with each included article, and the reduced data was collected in one place for further analysis. As Alhojailan [34] suggests, the data was displayed in several different ways to ensure that an in-depth understanding of the data could be achieved. These included (a) a large Word document table that included all

selected article excerpts or summaries (i.e., the reduced data), (b) thematic tables with supportive evidence noted, and (c) a concept map using the Obsidian platform.

The large Word document table containing all relevant barrier-related selected passages was reviewed independently by the team of researchers who identified overarching themes, and in a form of coding, the team pulled supportive evidence from the shared table into their own thematically categorized or coded working documents. Where the snippet of text pulled by the original researcher during data reduction did not provide enough context or needed clarity, the original articles were reviewed by other team members to clarify. This process allowed for inter-rater reliability to be assessed in several key ways. First, the content in the main group document with all selected passages was compared to what was drawn from there into the individual thematic analysis documents of each team member. Second, individual documents were compared to one another to see which excerpts were pulled by each researcher and how they were organized thematically. Finally, a second researcher reviewed a portion of the original articles to ensure a high percentage of relevant passages were in fact pulled to the larger shared table, suggesting little relevant information was lost in data reduction. In each case, high inter-rater reliability was demonstrated.

The majority of themes found (22 out of the 29) were initially identified and titled (i.e., coded) in conceptually similar ways by multiple reviewers (e.g., distrust of government, trust/fear, lack of trust). Where thematic titles were not clearly the same, further analysis and comparison of themes and the supportive evidence noted revealed a high degree of overlap between what researchers had noted, suggesting high inter-rater reliability despite the variance in titles used. Each researcher had not only pulled similarly sparse excerpts but had also consistently pulled the same excerpts to represent the same theme, further suggesting high inter-rater reliability. The unique themes identified by only one researcher, upon further review and reflection by the rest of the research team, were unanimously acknowledged as existing in the literature. Finally, guided by Ryan and Bernard's [35] suggestion to explore the unmarked sections (i.e., unselected passages or data thrown out in data reduction) for additional themes, a second researcher explored a random selection of articles (approximately 25 % of all articles reviewed) to identify any additional excerpts for inclusion that were not identified by the first researcher. Few additional relevant passages were identified by the second reviewer (i.e., less than ~15 %), suggesting a high level of inter-rater reliability between researchers and that further duplication of work likely would have provided limited returns on investment.

## 3. Results

Based on our review of the literature using the methodology described above, we identified the following five major themes discussed in this section: (1) the process is confusing and difficult to navigate; (2) exclusion from, or lack of access to, political power and disaster processes; (3) discrimination barriers; and (4) issues in the funding aid ecosystem (e.g., system failure, operations failure, procedural failure); and (5) politicians and media having an inordinate amount of control over the flow of resources and recovery processes. Each theme or issue is discussed below and noted with a "(number and letter)" to ease access and facilitate understanding of the organization of the discussion (see Table 1 for an overview). Additionally, many causes and examples of the themes identified will

**Table 1**
Overview of recovery barriers identified.

| |
|---|
| **1 The recovery process is confusing and difficult to navigate** |
| 1a excessive or burdensome paperwork |
| 1b navigating bureaucracies is difficult |
| **2 Exclusion from/lack of access to political power and processes** |
| 2a language and literacy limiting access |
| 2b difficulty accessing and using technology |
| 2c exclusion due to lack of transportation |
| 2d work duties including childcare limiting one's ability to fully participate |
| 2e geographic isolation from resources |
| 2f failure to account for cultural differences in help seeking behaviors |
| 2g lack of capacity at local levels |
| **3 Exclusion due to discrimination** |
| 3a socioeconomic class discrimination |
| 3b racial discrimination |
| 3c discrimination based on gender or sexuality |
| 3d fear of government and/or lack of trust |
| **4 Issues in the funding aid ecosystem** |
| 4a design of programs |
| 4b exclusion and/or prioritization of homeowners over renters |
| 4c systemic operational and procedural failures |
| 4d staff lacking competency and knowledge |
| 4e delays in system responses, communication, and disbursement of aid |
| 4f mismatch in timelines between program delivery and the amount of time individuals can reasonably survive on existing funds, family, and friends |
| 4g lack of transparent guidelines |
| 4h staff morale |
| 4i lack of staff diversity |
| **5 Politics and the media play an inordinate role in recovery outcomes** |
| 5a perceptions of viability |
| 5b political officials pushing their own agendas not equitable recovery |

also be noted with a letter, e.g., "(a)", again to help the reader navigate what are somewhat dense, example-rich passages. It is worth noting that none of the issues noted here exist in isolation, and, in fact, they all likely interact in synergistic ways, compounding the problem.

### 3.1. Theme 1: The process is confusing and difficult to navigate

One of the most prominent core themes identified here was that often the **processes and applications for disaster aid can be confusing and difficult to navigate** (1) [6,14,23,36–46]. These processes require significant "knowledge, skills and time" to navigate [37, p. 33], familiarity with working with government, and diverse forms of social capital to ensure informed navigation of these assistance programs [41].

Often, these disaster aid processes and applications are confusing because they are fraught with **excessive or burdensome paperwork** (1a). The literature highlights extensive paperwork is needed to verify ownership, which prevents people, especially those who've recently lost their homes, from applying for assistance due to a lack of required paperwork or documents (titles, birth certificates, marriage licenses, etc.), as well as other documentation issues that could only be resolved with legal assistance [5,6,23,47–50]. This appears to arise out of an organizational need to be good stewards of federal funds and protect them against fraud, though this anti-fraud lens has created barriers to recovery [8,51–53].

Another commonly cited explanation for this general confusion, especially for BIPOC populations, was the challenge of **navigating bureaucracies** (1b). This seems to emanate from (a) a general lack of understanding or experience dealing with the government, the Federal Emergency Management Agency (FEMA), and programs such as the Small Business Administration (SBA) [41,44,47,54]. These challenges of navigating bureaucracies seem particularly salient if a follow-up contact is required to verify aid eligibility [41], appeal an insurance or other repair claim [40], or if a series of applications is required, as is the case sometimes with individuals having to apply for SBA loans and being rejected to then be eligible for other forms of aid [38,40].

As touch points with different programs and their disparate requirements increase, so too does the level of confusion for individuals [5,10]. In at least one case, having to interact with several organizations with different requirements led to confusion and ultimately mistrust for a Southeast Asian community after Hurricane Katrina [5]. The barriers created by this (b) multi-stage application process and/or (c) piecemeal recovery efforts are that many people might not seek additional aid that they are eligible for due to frustration with the initial failed attempt. Furthermore, some people may terminate their pursuit of aid prematurely because of the confusion created by (d) poorly crafted communications. Examples of this are plentiful and include communications citing "deferred maintenance" or "insufficient damage" without defining these terms [7,14], determination letters that deem applicants "ineligible" without clearly articulating that they may still qualify for other forms of aid [40,55], or appeals for a determination [51]. Some authors suggest that these forms of poor communication reflect FEMA's "penchant for obfuscating available benefits and assistance" [55, p. 46]. Based on the literature reviewed, without clear and transparent communication and with the heightened intensity of immediate need, participants are often forced to make life-altering decisions hastily without all the facts, which ultimately results in outcomes that are frequently not in their best interest.

In addition to the challenges that arise from an unfamiliarity with programs, multi-stage processes, the involvement of many organizations and programs, and poor communication mentioned above, the literature suggests that there are several other factors contributing to this confusion and difficulty navigating programs. These include: (e) the level of complexity of both applications [4,56] and the service delivery system [6]; (f) confusing or unclear eligibility criteria [37,57]; (g) which is subject to repeated changes [42], as are aid distribution sites, which have been known to change locations hours beforehand [58]. Furthermore, as possible reasons for the confusion and difficulty navigating the process, the literature mentions participants (h) "getting the runaround"; (i) a lack of information, dated or even conflicting information [6,10]; (j) a scarcity of returned phone calls [59]; (k) the large number of organizations involved in recovery, often at different levels (state, local, etc.), each with different and sometimes conflicting requirements or goals [5,10,60]; and (l) a lack of public training [61]. A final topic in the literature is (m) frequent changes to deadlines and grant amounts. In one case, this led to early applicants being functionally penalized as funding amounts increased over time from $30,000 to $100,000, leaving those who applied early at a disadvantage [42]. Another report quantifying the number of changes in the Build It Back program in New York City after Hurricane Sandy shows that "between August 2013 and July 2014, HRO [Mayor's Office of Housing Recovery Operations] made more than 100 procedural changes to the Single Family Program" [59, p. 15]. This report demonstrates the extent to which changes in recovery programs often occur, and also offers a possible explanation for why people are not getting clear answers from staff operating these programs, which is addressed in more detail in subsequent sections (see Issues in Funding Aid Ecosystem).

### 3.2. Theme 2: Exclusion from, or lack of access to, political power and disaster processes

Being **excluded from or lacking access to political power and decision-making processes** was the second theme that emerged from the literature (2) [3,14,16,23,37,38,44,46,50,62–67]. Several examples of this theme include: (a) voter suppression and census undercounting [14]; (b) minorities and immigrants not participating in government decision-making processes due to discrimination, which leads to needs being overlooked [44]; (c) the exclusion of the elderly from decision-making processes [62]; and (d) asymmetrical political power leading to minority communities receiving less disaster aid due to disinvestment practices [23]. In several instances, it appears to be a deliberate exclusion from these processes [68]. Additionally, some communities, such as Indigenous tribes that have not received federal recognition, paradoxically due to their inability to present extensive Eurocentric documents, are simply not eligible for certain forms of aid [64]. In one reported instance where these communities managed to secure other forms of aid through nearly a decade and a half of planning and efforts, these funds and processes were quickly hijacked, rebranded, and utilized for other purposes not in line with the initial aims set out by the community themselves [65].

Being excluded from political processes has particularly salient implications for historically marginalized communities, as a later theme about politicians having undue influence on recovery will further highlight. For example, scholars investigating housing issues following major disasters have suggested that "uneven distribution of assistance is a function of political power and the ability to influence decision-making processes" [66, p. 535]. Furthermore, it's been demonstrated that communities able to politically advocate for themselves have been able to secure resources in the form of buyouts, whereas communities with less political voice might not be able to get such buyouts [50].

While there are instances where minority communities have successfully advocated for a voice at the table and even had their voices incorporated into a recovery plan, these seem to be an anomaly rather than the norm. One such instance was the New Orleans East Vietnamese community, which managed, with substantial advocacy efforts led by a local church, to secure a place at the table. They were able to use that opportunity to ensure the viability of their community, which was largely planned to be green space after Katrina [68,69]. It seems that this exclusion from, or lack of access to, political power and decision-making processes occurs for various reasons, some of which include: (a) a lack of effort to include communities in decision-making; (b) active attempts to exclude community voices [65,68]; and (c) communities choosing not to engage because of deep-seated mistrust of government entities [4,51].

For some limited English proficiency (LEP) and minority communities, research demonstrates that **language and literacy barriers limit access** to these important recovery processes (2a) [5,15,23,41,62,68]. Barriers for LEP communities include: (a) utter absence of application instructions in languages other than English [68]; (b) worries about drawing attention to the immigration status of themselves or family and friends [18,23,70]; and (c) limited ability to communicate with supporting organizations, first responders, and emergency providers [62]. Additionally (d) not having multiple forms of communication (e.g., both print and television), in several languages has been found to be a barrier [16]. This is a particularly concerning issue for individuals, LEP or otherwise, that are illiterate, especially in instances like that highlighted by Muñiz [18], where evacuation and weather advisories in non-written forms are only provided in English. Having only print material, without providing other forms of communication or support, also impacts the ability of illiterate individuals to access disaster recovery information [5]. The literature suggests these exclusions due to language are caused by a "lack of outreach, inadequate translation services" [23, p. 9] and a lack of forms and directions in other languages [68]. Lack of outreach may also be of particular salience in explaining why this exclusion occurs, given the cultural differences noted by Nguyen and Salvesen [5] in help-seeking behaviors.

Six other sub-themes were identified, one of which was exclusion due to **difficulty accessing and using technology** (2b) [44,53,71,72], and the disconnect between messaging channels used by emergency managers (i.e., technology) and those used by historically marginalized communities (i.e., trusted personal networks) [5,41]. Another was exclusion due to **lack of transportation** (2c), which not only prevents people from accessing disaster centers but also prevents aid from arriving and people going to work, which in turn, prevents them from being able to recover [4,23,38,41,58,62]. Lack of transportation prevents people from being able to apply for and maintain aid during the application process, as well as stabilize their income. Additional sub-themes include: **work duties including childcare limiting one's ability to fully participate** in the recovery process (2d) [38,58,73,74]; relative **geographic isolation from resources** (2e) [58,75]; including banks [76] and internet connectivity [64]; **failure to account for cultural differences** in help seeking behaviors (2f) [5,68]; and a **lack of capacity at local levels** limiting access to available resources, leaving rural communities especially limited in their ability to take full advantage of post-disaster resources (2g) [19,37,46,49,67,77,78].

Further literary analysis for the sub-theme of lack of capacity, specifically a lack of organizational and municipal capacity, highlights human capital, social capital, rurality, and the political makeup embodied in local government as important contributing factors to this issue. Some examples include (a) a limited ability to set up Volunteer Reception Centers (VRCs), which limits fund matching [19]; (b) staff lacking grant-writing skills [37]; (c) communities lacking pre-event planning, relationships, recovery training; and (d) not having a sufficient number of organizations or the infrastructure needed to accept and distribute funds [49,77].

The literature reflects the benefits of making use of community-based organizations to reach marginalized communities but also points to the fact that these organizations also often lack capacity themselves in terms of staffing and resources [37,67]. Lack of social capital is identified as a capacity barrier to recovery, with those living in rural areas being especially vulnerable to this when inconsistent impacts, failed disaster response, or displacement erode the historically strong social capital of some rural areas [49,79,80]. Interestingly, findings from Rumbach et al. [80] highlight the unique capacity and scaling-up challenges that small governments face through their comparative analysis of several geographically proximate communities of similar size with differing political makeups (e.g., liberal versus conservative) embodied and reflected in differing local government sizes. In their comparative analysis, Rumbach et al. [80] show that the larger local government engaged the communities better, was more trusted by the residents before and after the disaster, had more residents access disaster recovery centers, and had a bigger yearly operating budget. This meant that they had greater capacity when disaster struck and were more prepared to deal with the complexities of recovery than the nearby county with a smaller government.

### *3.3. Theme 3: Exclusion due to discrimination*

While part of the literature reviewed discussed or demonstrated systemic biases, many more couched the conversation within various forms of discrimination. In fact, the most frequently identified barrier to historically marginalized groups accessing disaster finances and resources in this review was **discrimination** (3). The extant literature primarily centers on discrimination based on race and class, while gender, sexuality, and disability are mentioned rarely, if at all. These are discussed separately but are intersectional issues that may in fact be inseparably connected, especially given the high rates of overlap between the impoverished and those who have been historically marginalized because of their race [7]. One particular type of discrimination found in the literature was **discrimination based on socioeconomic class** (3a) [15,17,47,75,81–84]. This type of discrimination appears in the literature on several fronts, including: (a) in buyout programs that have various nuanced barriers to participation for low-income individuals [84];

(b) around credit issues, including credit worthiness being utilized in the allocation of disaster loans and grants, access to credit [47, 81–83], and discouraged lending [75]. Another area where this is prevalent is in (c) the differences between aid received by homeowners compared to renters. A few stark examples of what could be considered socioeconomic class discrimination are highlighted by Domingue and Emrich [15], who note that counties with higher percentages of people who were unemployed or who were Social Security beneficiaries were 2909 % and 5972 % less likely to get high levels of funding, respectively, compared to counties with low levels of said groups. While socioeconomic class discrimination is a factor, the issue is multilayered and intersectional, with some research demonstrating funding discrepancies follow minorities despite their class status [17], suggesting race is a more important factor than class.

**Racial discrimination** (3b) was prevalent throughout the literature, including: (a) a lack of representation and opportunity to participate in government decision-making as previously noted [44,73]; and (b) a lower approval rate for Small Business Administration (SBA) loans in areas with a high percentage of minorities [23]. Other examples of racial discrimination include, (c) requiring eurocentric documents as noted by Kelley and cited in Harrynanan [64], while others suggest (d) gentrification, defined here as pushing out long-standing Black residents [85]; and (e) a decrease in assistance in areas with higher populations of BIPOCs [15,39,41]. One study demonstrated that counties with higher percentages of BIPOC populations got significantly less recovery money [15]. Strikingly, this study showed that counties with fewer Native Americans, Blacks, or Hispanics were 1316 %, 1032 %, and 825 % more likely to get high levels of funding, respectively, compared to counties that had higher percentages of these populations [15].

**Discrimination based on gender or sexuality** (3c) arises in the failure to recognize same-sex relationships, which occurred during Katrina and resulted in unequal access to disaster recovery resources [86]. Furthermore, women have unique perspectives [87] that do not seem to be well accounted for in the literature, despite them taking on unique roles in disasters and experiencing unique vulnerabilities such as limited access to resources due to gendered responsibilities [73]. This is particularly true for those with multiple historically marginalized identities [73]. Gender is discussed in some detail in terms of how it impacts response, but less so in terms of how it impacts recovery or availability of resources [86]. Generally, there is a dearth of literature around not only sexuality and gender [86] but also disability [62], which in itself may be a form of discrimination [86].

One poignant point made about the results of discrimination is the **fear of government and/or the lack of trust** many historically marginalized groups experience in the recovery process (3d). Several instances where this lack of trust has been highlighted in the literature include: mistrust between volunteers in predominantly Black communities and those in command of the recovery who were predominantly White; mistrust between residents and buyout program officials; and mistrust between program managers and low-income and elderly communities [16,19,50]. According to Grube et al. [41], "the literature highlights issues around … the mistrust of officials as possible explanations for who applies for and receives post-disaster assistance" (p. 578). The fear some minorities and immigrants have to apply for assistance, ask for help, or use available programs arises due to a fear of deportation, detention, or denial of residency for them or their loved ones. These are legitimate concerns, given that immigrants and their families are in fact routinely subject to surveillance, detention, and even deportation [38,41,62,66,67,88]. This fear to engage with programs exists regardless of an individual's actual immigration status, as many are still fearful to draw attention to others in their community, including family and friends [18,70]. Of note, one study found that among those who are "potentially undocumented," 38 % did not seek aid when needed since they did not want to draw attention to the immigration status of themselves or their family [4]. This is partially caused and certainly exacerbated by (a) past failures and historic government corruption [89], (b) rumors including immigration enforcement standing outside aid offices or tracking those who apply for aid [18,66], (c) aid workers in uniform who resemble the police force and government authority that so many have rightly come to fear, (d) lack of cultural competence, and (e) failures to approach these communities through trusted messengers [90]. This wariness to engage with disaster recovery programs as a response to present and past mistreatment is helpful in elucidating why minorities apply for aid at much lower rates, despite their need and eligibility for said aid [4,38]. Perhaps most importantly, this fear and distrust persist even with considerable outreach efforts [37].

### *3.4. Theme 4: Issues in the funding aid ecosystem*

The fourth identified theme was **"issues in the** funding **aid ecosystem"** (4) (e.g., language barriers, discriminatory programs, etc.). We have chosen to refer to these programs as the "funding aid ecosystem" to reflect that although FEMA is a focal point, several programs and policies come into play here. Although these programs are at times disparate, they interact at key touchpoints, representing a system of funding aid rather than a single homogenous source. It should be noted that many, if not all of the previously noted themes and sub-themes could be couched under this issue as well. However, it is important to discuss it under its own heading to elucidate issues not previously captured, and given this is a synthesis of literature, to capture how authors of the reviewed articles were framing these issues. This theme manifests in the literature as weaknesses in program design and operational or procedural failures such as staff failing to perform job duties [23,40,43,58,59,91]. As mentioned previously, historically marginalized groups experience these barriers to accessing disaster funds and resources disproportionately [8,16,53]. The identification of this barrier is particularly important because, in many ways, it may help link and explain large parts of the previously mentioned barriers.

One sub-theme of this overarching topic of issues in the funding aid ecosystem was the **design of disaster relief programs** (4a), which can be a barrier to access as it can result in the confusion, exclusion, delays, and discrimination previously mentioned. Examples found in this review of the literature include programmatic features that limit historically marginalized populations' access to disaster resources including: (a) complex processes for submission and appeals [47]; (b) unclear guidelines [14]; and (c) the **exclusion and/or prioritization of homeowners over renters,** which, due to its prevalence in the literature, represents its own sub-theme (4b) [17,23, 37,43,66,83,88,92–97]. In a study investigating FEMA's public assistance program, it found that counties with a low percentage of renters and mobile homes were 355 % and 186 % more likely to receive higher levels of funding, respectively, compared to counties with high percentages of rental units or mobile homes [15]. Further findings indicated that counties with lower gross rent were 83.23

% less likely to get high levels of funding compared to counties with high gross rent, further supporting the observation that there are clear differences in funding along socioeconomic lines. Other studies expounded on this theme, and highlighted the unique challenges faced by renters post-disaster, including increased costs for rent, the comparatively slow recovery of rental units, and the illegal eviction of renters that occurs as a result of landlords trying to capitalize on inflated housing markets [55,98]. The preferential treatment of homeowners can be seen in the amount of assistance funds distributed, the timeliness of assistance dispersal, and the lack of renter representation in recovery plans [23,89,99,100].

Another example of program design issues highlighted in the literature suggested that the biggest barrier to timely recovery identified by experts was (d) "undefined roles and responsibilities" [101]. Other examples of programmatic barriers include: (e) communication and application platforms that are predominantly online, limiting access to the elderly as well as immigrants [5,44], including those with limited broadband coverage [72]; and (f) a failure to include a broad array of stakeholders in the decision-making process [46]. Finally, (g) prioritizing highly valued assets in the distribution of aid [102], which leads to aid following wealth, not need, as allocations are dependent on the dollar amounts of damage [39], functionally and monetarily favoring the million-dollar getaway mansion on the coast over the fisherman's storm-torn stilted home.

Another related sub-theme was that of **systemic operational and procedural failures** (4c). Examples of this include (a) a demonstrable bias [41], with negative funding discrepancies following minorities despite their class [17], and (b) flood maps being outdated, leading to decreased rates of insurance [12]. Additional examples of this theme of operational or procedural failures include: (c) damage assessments only focusing on structural damage, ignoring other types of damage such as "loss of household contents, … disruption to income and/or psycho-social impacts" [58, p. 6]; (d) application of a "shared household rule", which has created a barrier to recovery funding for those sharing a dwelling [55]; and (e) FEMA's overuse of "deferred maintenance" to deny assistance, despite a lack of clarity, operationalization, or criteria around the use of the term [7,9,103]. Other examples found in the literature include (f) inequitable recovery rates influenced by different requirements used by different federal agencies [16]; and (g) federal programs historically taking a response-oriented approach, ignoring pre-existing vulnerabilities [37] [although, in more recent years, this has begun to be addressed with such frameworks as the National Disaster Recovery Framework (NDRF) [104] and programs like Building Resilient Infrastructure and Communities (BRIC) [105].

Another widely cited sub-theme was **staff lacking competency and knowledge** of both internal systems and procedures and of disaster-impacted areas and communities (4d) [6,8,12,37,40,41,58,88,97]. Examples of this include: (a) FEMA staff underestimating repair costs because they lacked familiarity with building codes and techniques [37], (b) administrative errors leading to thousands of applications being stuck in review years later [51,88]; (c) staff giving applicants incomplete or incorrect information [40]; (d) staff making blanket assumptions about the legal status of Latinos which led to neglect, discouragement, and even denial of aid to legal residents entitled to said aid [18]; (e) and paperwork being lost repeatedly leading to the all too common message, "we are unable to locate your paperwork" becoming a mainstream "mantra of frustration" [43, p. 107;58].

Some possible explanations for these apparent "incompetencies" found in this review include: (a) programs being run, sometimes almost entirely, by inexperienced and newly hired staff who receive abbreviated or no training at all [40,91]; (b) inadequate staffing [6]; and (c) the inability of staff to maintain awareness of the many rules and regulations [40,103]. For perspective, one program has more than 250 specific policies [103], which is compounded by the fact that these policies are ever-changing, in some cases over 100 times in less than a year, as previously highlighted [40]. The U.S. Government Accountability Office (GAO) [40] further highlights the extent of some of these issues, showing that 90 % of Disaster Recovery Center staff after Hurricane Irma and nearly all the staff in the Hurricane Maria response were newly hired trainees. This sheds some light on how confusion and unfamiliarity with the system at the staff level results in the same outcome at the individual recipient level, with confused staff being functionally unable to provide clear answers that they themselves do not have. While new hires are necessary to deal with the need for increased capacity after disasters, the need for consistency in programming and procedures is clear. What this reveals is that whoever happens to be assisting with a case has a disproportionate impact on the outcome, rather than policy or eligibility and procedure, which should be what guides eligibility determinations.

These repeated failures and inability to train staff to ensure levels of competence, including cultural competence, lead to: (a) delays [59]; (b) decreased accessibility to services [6]; and (c) biased damage assessments, which lead to historically marginalized communities receiving lower damage estimates and less assistance [23,41]. Additionally, this leads to (d) decreases in mental health [4,58] and (e) decreased quality of applicant interactions [6,40], which is amplified in importance by the findings of Sadri et al. [106], which found the quality and relevance of service were more important in reducing the time of recovery than the number of emergency managers involved.

Another barrier mentioned frequently was lengthy **delays in system responses, communication, and disbursement of aid** (4e), which occur often [43,45,48,76] and last for months [58,107], if not years, further delaying the arrival of critical funds for the community [51,107,108]. Recent literature suggests the top causes of delays, according to experts who rated a predefined list of 62 items, are (a) "undefined roles and responsibilities", (b) "improper physical development patterns and rules", and (c) "slow and improper decision-making" [101, p. 10]. Other literature traces these delays to: (d) administrative errors [51]; (e) inflexible and bureaucratic processes [91]; (f) confusing forms and requirements; (g) lack of returned phone calls; (h) resubmissions and reassessments [59,91]; and (i) loss of revenue base, which can prohibit starting to rebuild [91]. Additionally, delays are also being caused by (j) the lack of reliable digital access to shared case management files resulting in duplication of work [91]; (k) the sheer number of claims; (l) pending insurance claims paired with poor communication about alternate avenues of aid [55]; (m) the need to protect against fraud [53]; and (n) the related requirements to establish an address, "identity," and only one head of household [43,86,109]. This latter factor can become very challenging in homes where couples are separated [43] or non-heteronormative [109]. Delays disproportionately disadvantage marginalized groups in the race to recovery, stalling their recovery process [4,16,43], in some cases,

indefinitely [51,110]. This becomes a real issue as one study found "long waiting times to process applications determined recovery rates" (p. 6), forcing individuals waiting on FEMA to have to live "in precarious housing which compromised their well-being and wider recovery" [58, p. 6]. Such unrepaired damages may lead to further damages and, thereby, further depreciation of capital [81]. In at least one case, delays led to individuals being angry as they observed the inequalities in funding distribution and the timing of said funding [58], which likely had significant implications for future trust in the government and willingness to apply for aid, as highlighted in prior sections.

The significance and implications of delays for the recovery of historically marginalized communities are further highlighted by another closely related sub-theme, which was a **mismatch in timelines** between program delivery and the amount of time individuals could reasonably survive on existing funds, family, and friends (4f). This mismatch in timelines introduces a number of issues, which can be summarized in the idea that when delays occur without significant floating funds (i.e., more than the commonly recommended 3–6 months' worth of savings for a rainy day), you will sink [53,60,84,111,112]. Reid [53] first brought attention to this with the idea of "temporal domination" and the concept of "resource time" (i.e., the time before your resources are diminished to a level that basic needs cannot be met). A similar concept is also noted by Belury [111] called "poco a poco" or "little by little". These concepts, in sum, highlight that in many cases, this mismatch in timelines and lack of aid forces individuals to make decisions based on immediate pressing needs, which may result in less than ideal outcomes for their future recovery. Examples of this conundrum include individuals (a) accepting less than ideal buyout offers because they could not afford the extra cost and time involved in getting another appraisal [60]; and (b) parents having to choose between repairing moldy sheetrock in their asthmatic child's room versus putting money towards flood repairs on a family vehicle so they can get to work and continue to put food on their family's table [111]. Other examples include (c) families turning to predatory lenders when other options are not accessible, jeopardizing their long-term financial stability to satisfy immediate pressing needs [111], and (d) in another case, due to a delayed pending appeal for rental assistance, a family was forced to put money towards rent rather than towards immunizations required for their children to attend school [53]. In this particular case, the added stress and strain led to domestic violence and ultimately the incarceration of the partner [53]. A final poignant example with concrete statistics of this juxtaposition of immediate needs demanding decisions that hurt long-term outcomes, largely due to delays and programmatic policies, is highlighted by Baker et al. [60], where they explore buyout programs. This final example (e) demonstrated that 75 % of individuals in the study's sample had to put insurance funds towards repairs on a home that would be demolished, along with their chances of having enough to pay for a new home when the buyout program funds finally came through. This was due to "double dipping policies" in the Stafford Act, which prohibit federal funds from paying for things that have or will be paid for by another source, and because program delivery in this case took, on average, over a year while the average time individuals could float on family, friends, and funds was about six months before their funds were exhausted and relationships strained. It is important to note that having savings for three to six months is commonly considered best practice for saving for hard times, suggesting these individuals were likely reasonably prepared, just not for the excessive delays that ensued and required them to stay in limbo for longer than their savings and social capital allowed. It's also worth noting that in other cases, buyout program delivery has been shown to take five to ten years [84], which, if data were available for such instances, would likely show similar trends to those noted here, if not more exaggerated.

Elliot and Pais [112] suggest that people with financial savings float on those funds at post-storm rentals such as an apartment or hotel, while poorer people rely more heavily on family. Nelson et al. [84] highlight a similar finding, suggesting lower-income residents are more impacted by long wait times because they cannot pay for temporary housing. This is compounded by higher rates of job loss, like those highlighted in Elliot and Pais [112], which show Black workers were four times more likely to lose their job than Whites after a disaster. One's ability to wait out delays could also be exacerbated by loss of income due to increased work absences related to poor mental health after a disaster, which has been shown to be as high as 18.4 days per year, resulting in a loss of $4198.33 for individuals annually, not an insignificant number, especially if one is below or near the poverty line [113]. Missing work or losing your job after a storm certainly impacts one's financial cushion, and having that financial cushion seems to allow more privileged individuals to pay for temporary housing, giving them a second buffer as they still have family to fall back on if their floating funds have run out [53,112]. Those with what Reid [53] calls more "resource time" do not have to make immediate need-based decisions, which would likely lead to worse long-term recovery outcomes. These additional cushions (i.e., extra floating funds and a secondary buffer of being able to fall back on family and friends only after savings have run out) allow wealthier individuals to "float" through the frequent and lengthy delays, while poorer individuals are forced to make decisions that appease immediate needs but ultimately undermine their long-term recovery, ultimately forcing them off the precipice of precarity and into downward spirals of resource loss [60,111,112]. For additional unpacking of how delays act as a social stratification sieve, further widening the recovery gap between the "haves" and the "have nots", see our research brief [114].

As mentioned previously, systemic biases and outcomes abound in the literature. Based on the literature reviewed, it appears that one of the main avenues through which these systemic biases and personal agendas are allowed to flourish is through a **lack of transparent guidelines** or clear set procedures (4g), leaving just enough wiggle room for conscious and unconscious biases to operate undetected in a supposedly "color blind" system [8]. Some authors have even suggested that in FEMA's case, this lack of procedural guidelines exists precisely because they wish to avoid enforceable procedural rights [55]. This results in the pipeline of aid frequently being guided by things other than need.

One example showed a "lack of clarity surrounding the roles of home inspectors that invited racism, classism, and prejudice" and that "without clear procedure for each inspection, and oversight to ensure these procedures are being practiced, inspectors could introduce discrimination in the IHP process" [14, p. 229]. This was done by using the term "deferred maintenance", which was disproportionately used with low-income populations [14,23]. Unclear guidelines have also led to contractor abuses and incorrect prioritization of aid [23,63,66,89,107]. Damage inspectors can produce biased readings when their inspections are cursory, wage

incentives promote quick work, not thorough work, and data collection protocols lack clarity or transparency [23,103]. Other examples of this theme of lack of clear set procedures include the fate of affordable housing being "open to discussion" having "no previously agreed upon course of action" [63, p. 1632], and buyout programs with unclear and arbitrary decision-making criteria around the determination of the geographical boundaries for eligibility for aid [10]. A lack of transparency around decision-making in the application and appeal processes is also noted as a barrier [7,10,50,51,60,115]. This can create bitterness in residents that can persist throughout the aid process [60], potentially impacting their further engagement and pursuit of aid.

Residents are not the only ones experiencing confusion; policymakers are uncertain about the logistics of compensation programs, which can cause significant delays in the disbursement of support [107]. It's been noted that "most programmes lacked clear guidelines for identifying areas of greatest need and prioritizing aid accordingly," allowing for political agendas to direct funds for reasons other than need [66, p. 557]. It is crucial to highlight the juxtaposition that exists within this barrier: that community members need clear, consistent protocols but also flexibility [18,84]. The fluidity necessary to meet the needs of varied communities and community members is often met with trade-offs, as allowing flexibility in the local governance of funds also exposes them to the influence of local politics [21,107].

Another important sub-theme found in the literature that offers a unique perspective is the impact of **staff morale** on the overall process (4h). Disaster recovery staff experience both new and long-standing challenges that impact fatigue, morale, and overall mission success. Unfortunately, these negative staff experiences cascade downward and result in negative experiences of disaster survivors as customers (recipients) of disaster recovery programs [40,43]. As noted in the literature, "another unique quality of the staff is that they too have been traumatized by their own disaster experience, yet must of course listen continuously to survivor after survivor's description of their disaster experiences" [43, p. 116]. According to the GAO [40], customer (disaster survivor) satisfaction (with regard to their treatment and case management) is linked to employee satisfaction, noting issues such as inadequate training, fatigue, and low morale for Disaster Recovery Center employees [40]. In examining the Individuals and Household Program (IHP), the GAO found that even in ideal situations, staff cite being overburdened and underappreciated, inadequately prepared to meet both their own and survivors' needs in traumatic circumstances in both contexts of administrative interactions and interactions with disaster survivors [40].

The **lack of staff diversity** among disaster recovery agencies was frequently cited in the literature (4i), which acts as a barrier to recovery for communities whose demographics are not reflected in the staff and leadership of these programs. Writing on the ways agencies and private disaster relief failed the Latino community following Hurricane Katrina, Muñiz [18] suggests a lack of diversity among staff and volunteers led to a lack of assistance for Spanish-speaking survivors, some of whom had their needs overlooked because they were mistaken for recovery workers. Similarly, Williams and Jacobs [19] explore how the insularity of White emergency management professionals creates a barrier to engaging Black recovery volunteers and disaster survivors, which, in turn, results in barriers to assistance and aid. In their research on barriers to recovery among multiple Southeast Asian ethnic groups following Hurricane Katrina, Nguyen and Salvesen [5] highlight the importance of hiring staff who can speak the languages of those seeking recovery aid and who understand the cultural differences in help-seeking behavior, which was lacking in the Katrina recovery. In each case, a lack of staffing from different backgrounds meant that marginalized individuals seeking aid were left with obstacles to their recovery stemming from this mismatch in representation between those seeking aid and those dispensing it.

### *3.5. Theme 5: Politics and media play an inordinate role in recovery outcomes*

The fifth identified theme was that **politicians and the media play an inordinate role in recovery outcomes** via their ability to direct or restrict resource flows (5). Two sub-themes are represented here, the first being that narratives are created about communities by politicians and the media that impact **perceptions of viability** (5a) and, ultimately, the distribution of funds. The literature suggests these perceptions of viability, or closely related "deserving" or "worthy" victim narratives [8,43], are impacted by both political leaders [94] and the media [12].

Examples of this theme include Mayor Nagin declaring post-Katrina parts of Uptown New Orleans "immediate opportunity areas", which funneled resources into this community that ultimately helped it recover quickly [94, p. 411]. Communities such as the Lower Ninth Ward were not so fortunate, however, with the media uniformly describing the community, despite varying ranges of damage, as a place of 'total destruction', 'without a sound', 'nothing there to salvage' 'just going to have to be bulldozed' as cited in Green et al. [12, p. 324], which ultimately impacted the community's ability to recover. As Green et al. [12] highlight, "these descriptions of physical damage have been used to call into question the logic of rebuilding neighbourhoods" (p. 324) and ultimately have impacted community members' willingness to return and begin rebuilding, due to the rumors of their homes being doomed for bulldozing. Other literature suggests that "narratives create a fear of fraud that leads to extensive expenditures and hinders the ability to deliver aid to those who most need it" [8, p. 28]. In the case of the Lower Ninth post-Hurricane Betsy, these kinds of narratives led to the "explicit suggestion that giving money to the poor would lead to dependence," which led to aid being deemed "an unacceptable option" [8, pp. 32–33].

The literature highlights not only the adverse impacts of negative media attention but also the injurious impacts a lack of media attention can have [8,43,110]. Similarly, stigma and bias surrounding the viability of colonias (low-income communities along the U.S.-Mexico border) negatively impacted the execution of the FEMA IHP program post-Hurricane Dolly [14], and stigmas regarding public housing drove decision-making processes around rebuilding public housing in Galveston, Texas after Hurricane Ike in 2008 [63]. In each of these cases, perceptions of viability (or lack thereof) for recovery guided decisions about recovery aid in a way that disadvantaged people from historically marginalized groups.

As highlighted in the example above with Mayor Nagin, a second related sub-theme identified was that sometimes in disaster recovery, **political officials push their own agendas** (5b) rather than promote equitable recovery. This results in aid following

political interests rather than need. Often, the agendas being pushed are due to pressures from affluent stakeholders, which then leave the interests of historically marginalized communities neglected [116]. According to an analysis done by Garrett and Sobel [20], "nearly half of all FEMA disaster relief is explained by political influence rather than actual need" [20, p. 508]. Sobel and Leeson [21] found that "when government is in charge of disaster relief resources, political actors seeking private ends, such as re-election, face an irresistible incentive to cater to important geographic constituencies, which are not always the constituencies in most need of assistance" (p. 60). The unpredictability and political nature of such streams of funding are therefore not ideal for communities in genuine need of funding and resources [45], and this is particularly true for those typically excluded from these processes in the first place, as previously noted.

Another example suggests that states' roles in elections played a part in the amount of money communities received after a series of tornadoes, citing the $10.1 million a non-swing state received compared to the $150 million a swing state received after each was impacted by a tornado in the same year [117]. Green et al. [118] highlight that local, state, and national leadership post-Katrina gave preference to the interests of developers and large businesses over those of low-income, African American populations. When political officials seek to advance their own agendas, they can become barriers to recovery for communities rather than facilitators of recovery. A few key examples of this phenomenon include: (a) government officials hijacking a tribal resettlement initiative once funding was secured and erasing their efforts, voice, and vision for what that would look like [65]; and (b) officials declaring no funds were available for maintenance of the single road into the Isle de Jean Charles community, but then later heavily investing in said road and fishing piers once many tribal members had left [65]. Other examples include: (c) the New Orleans East Vietnamese community having to fight political officials for its survival [69]; (d) initiatives seeking to improve language access being hindered by political officials that lacked cultural understanding [90]; (e) policy makers preventing facilities that serve the working and very poor including the public schools, hospitals, and housing from being rebuilt [89,118]; (f) political corruption adding to the level of distrust communities already have towards government [89]; and (g) city and state officials pushing different programs and recovery agendas leading to confusion [10]. As Spader and Turnham [107] highlight, allowing more localized control and flexibility of funds seems like a good idea to ensure targeted and tailored use of said funds, but this can be a double-edged sword, as it exposes programs to the whims of local politics. It should be noted here that while this section focuses generally on more local contexts and examples, FEMA itself is not immune from political inputs that may differ from administration to administration.

This theme of political officials and the media having inordinate control over the flow of resources and impeding recovery for communities is particularly problematic for historically marginalized communities that have limited access to the political power structure, as previously highlighted, and who frequently find themselves on the battle front of the "viability," "deserving," and "worthy of aid" narratives fighting for the survival of themselves and their communities. This is even more salient for immigrants and their families, who are often on the extreme end of political exclusion and toxic narratives. The literature suggests that this willful, or potentially inadvertent, maneuvering of resources (which may be the case for the media) seems to be possible due to a lack of clear guidelines or specific programmatic charge to prioritize aid where it is most needed [9,66].

## 4. Discussion

The literature reviewed here provides a synthesis of the barriers that communities, particularly BIPOC communities, face when trying to access disaster finances and resources. The goal is to shed additional light on the issues at hand and some of their potential causes. The overarching themes found include a confusing process, exclusion, discrimination, failures within the ecosystem of aid, and the role politicians and media play in the distribution of resources or lack thereof, all of which have historically been issues and remain issues that need addressing in the recovery process (for a full break-down of themes and sub-themes, see Table 1). These themes identified align closely with what advocates, communities, and practitioners report as challenges in disaster recovery. While there is overlap between our findings and global literature in terms of recovery barriers identified (e.g., delays, distrust, unclear regulations, political pressures), our research largely expands the growing list of disaster recovery barriers previously uncovered in prior global reviews [22,119]. While the mechanics or causal factors behind the distinct barriers identified in this review may be somewhat unique to the U.S. context given the distinct policies, programs, and practices, it's likely that many of these issues are also experienced globally. Global resilience scholars have increasingly focused on the social factors of resilience (see Ref. [120,121]), and our review also shows that these social factors greatly influence community resilience, in particular trust, which is an overlooked infrastructure that takes significant time and effort to build. Preparedness, adaptation, mitigation, and recovery efforts would greatly benefit from increased investments in trust infrastructure with marginalized communities, particularly in what the literature calls "linking social capital" [120,122], which "connects regular citizens with those in power" [120, p. 259]. Disasters exacerbate power dynamics in existing power structures and inequities, and the issues found here in recovery outcomes are simply post-disaster manifestations of more deeply seated issues. Even though systemic failures are clearly present and demonstrable in the literature, it is challenging for vulnerable groups to make appeals as the process is lengthy, costly, and burdensome [8]. In such cases, Verchick [97] suggests "special government compensation should not depend on how popular, well-connected, or lawyered-up the injured community is" (p. 66).

In each case, the barriers noted here impact historically marginalized communities disproportionately, while leaving those with what Schwartz [123] called "immunity of the privileged" only minimally impaired by these barriers. This seems to be particularly true when it comes to delays, which appear to be a form of what Reid [53] called "temporal domination". Functionally, delays act as a social stratification sieve, propelling those with resources into recovery and pushing those without off the precipice of precarity into downward spirals of resource loss. This finding is particularly salient given the typical timelines of program delivery, often measured in years [51,84,88,107,108], while people's floating time is measured in months, if not days. Recent statistics show 61 % of Americans have less than six months of savings [124], and around half (46–51 %) have less than three months worth of savings [124,125].

Furthermore, 32 % cannot cover a $500 emergency expense with their savings, and nearly half of those cannot even cover a $100 emergency expense [126]. These differences in savings follow racial lines [125], and income patterns (i.e., those who make more can save more) [124]. These problematic delays, in part, seem to arise due to a need to prevent fraud and possibly an overzealous attempt to do so [8,53]. In instances where there has been fraud, given the evidence of high levels of confusion noted in this review, it seems nearly impossible to suss out whether these were cases of ill-intentioned manipulation of the system or simply reasonable misunderstandings related to program eligibility requirements and rules. With communities currently burdened with weaving together uncoordinated and piecemeal recovery resources, the latter seems highly probable.

Dr. Robert Bullard, the father of environmental justice, succinctly summarized the majority of what was found in this literature review, stating "money follows money, money follows power, money follows Whites" [2]. It is clear that disaster financing in the U.S. has historically been reactive, exclusive, and subject to the discretion of the few with power and resources. Resources beget resources, and the complexity of the ecosystem of aid reflects a nearly unnavigable process for many communities. Pre-existing power structures, including development pressures, have a substantial impact on the decisions that are made following a disaster. Additionally, research has highlighted the gendered nature of disaster aid and recovery and the potential for women, in particular, to have lower levels of access to disaster finances and resources because of oppressive experiences of social control [50]. Despite these differing experiences and outcomes based on one's gender, few studies have explored these topics through a gendered lens.

A small body of literature found positive outcomes (e.g. Ref. [15], or [68]) for historically marginalized communities in recovery, which may indicate changes are being made within the ecosystem of aid. However, it seems, based on our review of the last 25 years of literature, that the rate of change has been rather sluggish historically and appears to only move when prodded along by court cases and public backlash. Most recently, however, there have been changes within the ecosystem of aid brought about by the Justice40 initiative and the influx of funding it brought with it, which has the potential to be a catalyst to make lasting change and help address some of these long-standing issues noted in the recovery process [127,128]. Despite this recent momentum and focus on equity, issues will likely continue to exist, as the subsequent paragraph highlights.

The National Advisory Council to FEMA suggests that to improve upon current practices, emergency managers need to (a) understand equity, (b) "become equitable in every approach", and (c) become equitable in all outcomes [9, p. 11]. These can be conceptualized as three steps to reaching equity, with the first two being more functionally and feasibly achievable in the near future. While becoming equitable in all outcomes is certainly achievable, as demonstrated by the majority of measures in a recent study [15], it is likely to occur in a more distant future. This is because even with a "magic wand" that could somehow ensure that the entire ecosystem of aid understood equity and utilized this understanding in every approach, this would not erase the distrust that has been deeply entrenched through centuries of colonization, extractive practices, racism, and programs that have consistently delivered outcomes that are not equitable and have, in fact, typically disadvantaged those already historically marginalized. Some have noted that "participants' expectations for government agencies … [are] shaped by a history of perceived failures by multiple levels of government'' [10, p. 101], and this deep-seated mistrust has been shown to persist even with considerable outreach efforts [37], suggesting gaps in equitable outcomes would likely continue to exist due to past experiences and the lingering fears and mistrust generated by that history. This would likely prevent many from participating in programs long after equity was understood and utilized in every approach. If one conceptualizes trust through the lens of banking, it becomes clear that the ecosystem of aid is always coming to the table with a large trust deficit when approaching historically marginalized communities, and considerable effort and consistent trust deposits will have to be made over time just to get out of the "red", back to baseline, and then into the "black". This begs the question, can trust infrastructure be built at a rate that is asymmetrical to how the distrust was created in the first place? One can only hope and try.

### *4.1. Study limitations*

The review had several limitations, including: (a) one subgroup of the research team demographics were rather homogenous (b) an overreliance on backward citing (i.e., snowball sampling references); led to an imbalanced use of dated sources (i.e., >7 years old); (c) the vast breadth of the search did not allow in-depth analysis of all disaster-related literature; and (d) a lack of capacity did not allow every article to be fully reviewed by two or more researchers. These limitations are discussed in more detail below.

Our team consisted of 16 individuals: eight identified as females, eight as males, and five team members identified as minorities from underrepresented groups. The research team was loosely divided into two subgroups: a team of eight graduate students and a team of eight faculty and staff members at Tulane University. The graduate team was predominantly White and U.S.-born, which was a limitation. This is largely due to, and reflective of, the pool of graduate students at the host institution, a predominantly White, private institution. While significant recruitment efforts were made to recruit more diversely, our limited pool of candidates and their limited bandwidth for involvement greatly limited our ability to recruit a more diverse graduate team. Having a more diverse team overall, including more BIPOC members, throughout the process could have led to more insights or unique perspectives that may have been missed. That being said, the faculty and staff group was more diverse, and this may have partially compensated for this limitation.

The overreliance on backward citing was partially compensated for in a final wave of analysis that focused entirely on the most recent literature. While relevant literature certainly may have evaded capture, it is believed that thematic saturation was reached and that further exploration would have yielded few, if any, new themes. Thematic saturation was gauged by the high overlap of themes between the various waves of literature review and the limited new themes being added with the review of additional articles. It is obviously possible that with further exploration, additional themes could have been found or the framing of previously found themes could have slightly changed, but this is normal with thematic analyses as the slight reframing and revisions of themes could go on endlessly [129].

## 5. Conclusion

There were a few noteworthy gaps identified in the literature reviewed, including: (a) few studies explored gender differences in outcomes and experiences in recovery; (b) a general lack of concrete operable recommendations and a complete dearth of literature on any of these being tested; (c) a lack of explorations of barriers faced internally by the ecosystem of aid; and (d) a lack of studies taking a positive deviance lens where exceptional stories of success are explored to identify immediately operable strategies that could be replicated to amplify patterns of success. Noteworthy exceptions to the research gaps highlighted include the GAO [40,91] reports and the case of positive deviance found in the recovery of the New Orleans East Vietnamese community [68]. While this later case study offers helpful insights into what led to their success and demonstrates what is possible with strong community ties and leadership, it also serves as a shameful reminder of the literal fight for survival some communities must engage in on top of the normal efforts required to recover. No studies reviewed used a positive deviance lens to explore the ecosystem of aid, despite empirical evidence demonstrating that in some responses, programs have operated in equitable ways in certain areas of the response [15]. Future studies should further explore the success stories of both communities and the ecosystem of aid to more fully understand how these patterns of success can be amplified and replicated. Particular attention should also be paid in future studies to gender differences in terms of an individual's experience and outcomes in disaster recovery. Additionally, while this review focused on government-sponsored programs, there are other significant streams of funding available to help communities trying to recover. Future research could explore philanthropic, private, and government collaborations. In such future research endeavors researchers should follow principles outlined in the *Disaster studies manifesto* by Gaillard et al. [130], which outlines current concerns for disaster research, what needs to happen, and how to get there.

This review found several overarching themes, including confusion, exclusion, discrimination, and failures within the ecosystem of aid and the role politicians and media play in the distribution of resources or lack thereof. These findings echo the voices of communities and their advocates, and affirm the findings of other literature reviews that have explored similar topics both domestically and globally [22,23,45,54,119,131] while adding to the body of literature in unique ways. The authors conclude by encouraging researchers going forward to explore not just what is not working but feasible solutions to these complex challenges. The barriers to recovery are now well known; solutions, however, remain elusive and will remain so unless multidisciplinary cross-sectoral efforts make a more concerted effort to solve these issues.

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## CRediT authorship contribution statement

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

No data was used for the research described in the article.

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